

4.2.1a Evaluate a diabetic patient presenting with a foot wound at 3 levels: the patient as a whole, the affected foot or limb, and the infected wound. IDSA 2012 guidelines (Adapted)

4.2.1b Assess the affected limb and foot for arterial ischemia (strong, moderate), venous insufficiency, presence of protective sensation, and biomechanical problems. IDSA 2012 guidelines (Adapted)

Explanatory note: Biomechanical problems means anatomical and physiological disturbances of the foot. i.e., structural changes which happen in the bones, joints and muscles of the foot of diabetics and the changes in the blood circulation and nerve sensation of the foot of diabetics.

4.2.2. Classification of Diabetic foot

4.2.2a Assess the severity of any diabetic foot infection using the Infectious Diseases Society of America/ International Working Group on the Diabetic Foot Classification system. (Strong; Moderate) IWGDF 2015 (adopted)

(Refer to Annexure below for IDSA/IWGDF classification system -Table 1)

4.2.2b Do not use the Wagner classification system to assess the severity of a diabetic foot ulcer. (Adopted) (NICE 2015 Guidelines on Diabetic foot problems: prevention and management.)

4.2.3 Referral for Diabetic foot problems

4.2.3a Initially hospitalize all patients with a severe infection, selected patients with a moderate infection with complicating features (eg, severe peripheral arterial disease [PAD] or lack of home support), and any patient unable to comply with the required outpatient treatment regimen for psychological or social reasons. Also hospitalize patients who do not meet any of these criteria, but are failing to improve with outpatient therapy. IDSA 2012 guidelines (Adapted)

(Also refer to Table-3 and 4 in the Annexure below, for explanatory notes.)

4.2.3b Prior to being discharged, make sure that a patient with a DFI (Diabetic Foot Infection) is clinically stable; has had any urgently needed surgery performed; has achieved acceptable glycemic control; is able to manage (on his/her own or with help) at the designated discharge location; and has a well defined plan that includes an appropriate antibiotic regimen to which he/she will adhere, an off-loading scheme (if needed), specific wound care instructions, and appropriate outpatient follow-up. IDSA 2012 guidelines (Adapted)

4.3: DIABETIC FOOT INFECTIONS

4.3.1 Consider the possibility of infection occurring in any foot wound in a patient with diabetes. Evidence of infection generally includes classic signs of inflammation (redness, warmth, swelling, tenderness, or pain) or purulent secretions, but may also include additional or secondary signs (e.g., non-purulent secretions, friable or discoloured granulation tissue, undermining of wound edges, foul odour). IDSA 2012 Guidelines (Adapted)

4.3.2 Be aware of factors that increase the risk for diabetic foot infections (DFI) and especially consider infection when these factors are present; these include a wound for which the probe-to-bone (PTB) test is positive; an ulceration present for >30 days; a history of recurrent foot ulcers; a traumatic foot wound; the presence of peripheral vascular disease in the affected limb; a previous lower extremity amputation; loss of

protective sensation; the presence of renal insufficiency; or a history of walking barefoot. IDSA 2012 guidelines (Adapted)

4.3.3 Take plain radiographs of the affected foot of all patients presenting with a new Diabetic Foot Infection to look for bony abnormalities (deformity, destruction) as well as for soft tissue gas and radio-opaque foreign bodies. IDSA 2012 Guidelines (Adapted)

4.3.4 When and how to obtain culture from Diabetic foot patients?

4.3.4a For clinically uninfected wounds, do not collect a specimen for culture. IDSA 2012 guidelines (Adapted)

4.3.4b Send a specimen for culture that is from deep tissue, obtained by biopsy or curettage and after the wound has been cleansed and debrided. Avoid swab specimens, especially of inadequately debrided wounds, as they provide less accurate results. IDSA 2012 guidelines (Adapted)

Explanatory note: Wash the wound with saline and the surrounding skin with antiseptic solution before taking culture to avoid contamination of the specimen obtained for culture.

4.3.4c Do not obtain repeat cultures unless the patient is not clinically responding to treatment. IWDGF 2015 Guidance Document (Adapted)

Explanatory note - Expert consensus says that if the signs of inflammation do not subside even after 72 hours of starting treatment, then it should be considered that patient is not responding.

4.3.4d For infected wounds, send appropriately obtained specimens for culture prior to starting empiric antibiotic therapy, if possible. Cultures may be unnecessary for a mild infection in a patient who has not recently received antibiotic therapy. IDSA 2012 guidelines (Adapted)

4.3.5. Selection of Antibiotic and when should it be modified?

4.3.5a Do not treat clinically uninfected wounds with antibiotic therapy. IDSA 2012 Guidelines (Adapted)

4.3.5b Prescribe antibiotic therapy for all infected wounds but caution that this is often insufficient unless combined with appropriate wound care. IDSA 2012 guidelines (Adapted)

4.3.5c Base the route of therapy largely on infection severity. Prefer parenteral therapy for all severe, and some moderate DFLs, at least initially, with a switch to oral agents when the patient is systemically well and culture results are available. Use highly bioavailable oral antibiotics alone in most mild, and in many moderate, infections and topical therapy for selected mild superficial infections. IDSA 2012 guidelines (Adapted)

4.3.5d Select an empiric antibiotic regimen on the basis of the severity of the infection and the likely etiologic agent(s).

- a. For mild to moderate infections in patients who have not recently received antibiotic treatment, therapy just targeting aerobic gram-positive cocci (GPC) is sufficient.
- b. For most severe infections, start broad-spectrum empiric antibiotic therapy, pending culture results and antibiotic susceptibility data.
- c. Empiric therapy directed at *P. aeruginosa* is usually unnecessary except for patients with risk factors* for true infection with this organism.

d. Consider providing empiric therapy directed against MRSA in a patient with a prior history of MRSA infection; when the local prevalence** of MRSA colonization or infection is high; or if the infection is clinically severe.

IDSA 2012 guidelines (Adapted)

Explanatory notes:

* Risk factors for true infection with *Pseudomonas aeruginosa* include Immunocompromised status, Chronic Kidney Disease, warm climate and frequent exposure of foot to water.

**The local prevalence of MRSA (i.e., percentage of all *Staph. aureus* clinical isolates in that locale that are methicillin resistant) is high enough (perhaps 50% for a mild and 30% for a moderate soft tissue infection) that there is a reasonable probability of MRSA infection.

4.3.5e Give an initial antibiotic course for a soft tissue infection of about 1–2 weeks for mild infections and 2–3 weeks for moderate to severe infections. IDSA 2012 guidelines (Adapted)

4.3.5f Continue antibiotic therapy until, but not beyond, resolution of findings of infection, but not through complete healing of the wound. IDSA 2012 guidelines (Adapted)

4.3.5g Administer parenteral therapy initially for most severe infections and some moderate infections, with a switch to oral therapy when the infection is responding. (Strong; Low) IDSA 2012 guidelines (Adopted)

4.4: WOUND CARE

4.4.1 Clean ulcers regularly with clean water or saline*, debride them when possible in order to remove debris from the wound surface and dress them with a sterile, inert dressing in order to control excessive exudate and maintain a warm, moist environment in order to promote healing**. (Strong; Low)IWGDF 2015 (Adopted)

Explanatory note:

*Clean water is boiled cooled water (distilled water)

** Do not use hydrogen peroxide, EUSOL (Edinburgh University Solution), povidone iodine, chlorhexidine, etc

4.4.2 Select dressings principally on the basis of exudate control, comfort and cost. (Strong; Low) IWGDF 2015 (Adopted)

4.4.3 Do not use antimicrobial dressings with the goal of improving wound healing or preventing secondary infection. (Strong; Moderate) IWGDF 2015 (Adopted)

4.4.4 Do not offer the following to treat diabetic foot ulcers, unless as part of a clinical trial:

- Electrical stimulation therapy, autologous platelet-rich plasma gel, regenerative wound matrices and dalteparin.
- Growth factors (granulocyte colony-stimulating factor [G-CSF], platelet-derived growth factor [PDGF], epidermal growth factor [EGF] and transforming growth factor beta [TGF-β]).
- Hyperbaric oxygen therapy.

NICE clinical guideline NG19 (Adopted)